Binocular MIPI Image Acquisition

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Function Introduction

To achieve 3D environmental perception, robots typically incorporate sensors such as binocular cameras and Time-of-Flight (ToF) sensors. To reduce user costs associated with sensor adaptation and usage, TogetheROS.Bot encapsulates various commonly used sensors and abstracts them into the `hobot_sensor` module, supporting ROS standard image messages. When the configured sensor parameters do not match the connected camera, the program automatically adapts to the correct sensor type. Currently supported MIPI sensor types are as follows:

Type	Model	Specification	Supported Platform
Camera	SC132gs	130W	RDK X5, RDK X5 Module, RDK S100, RDK S100P

code repository：https://github.com/D-Robotics/hobot_mipi_cam.git

Supported Platforms

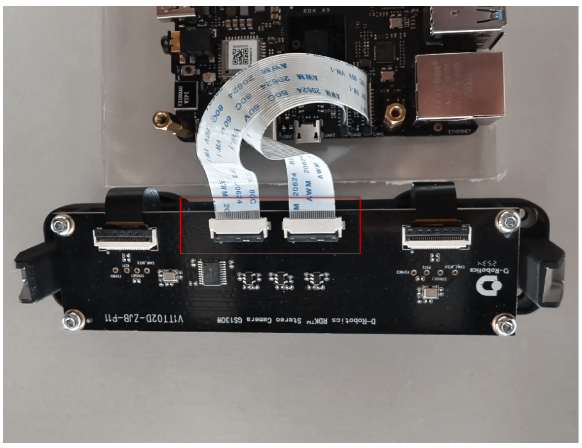
platform	Operating mode
RDK X5, RDK X5 Module	Ubuntu 22.04 (Humble)
RDK 100, RDK S100P	Ubuntu 22.04 (Humble)
RDK 100, RDK S100P	Ubuntu 22.04 (Humble)

Preparation

RDK Platform

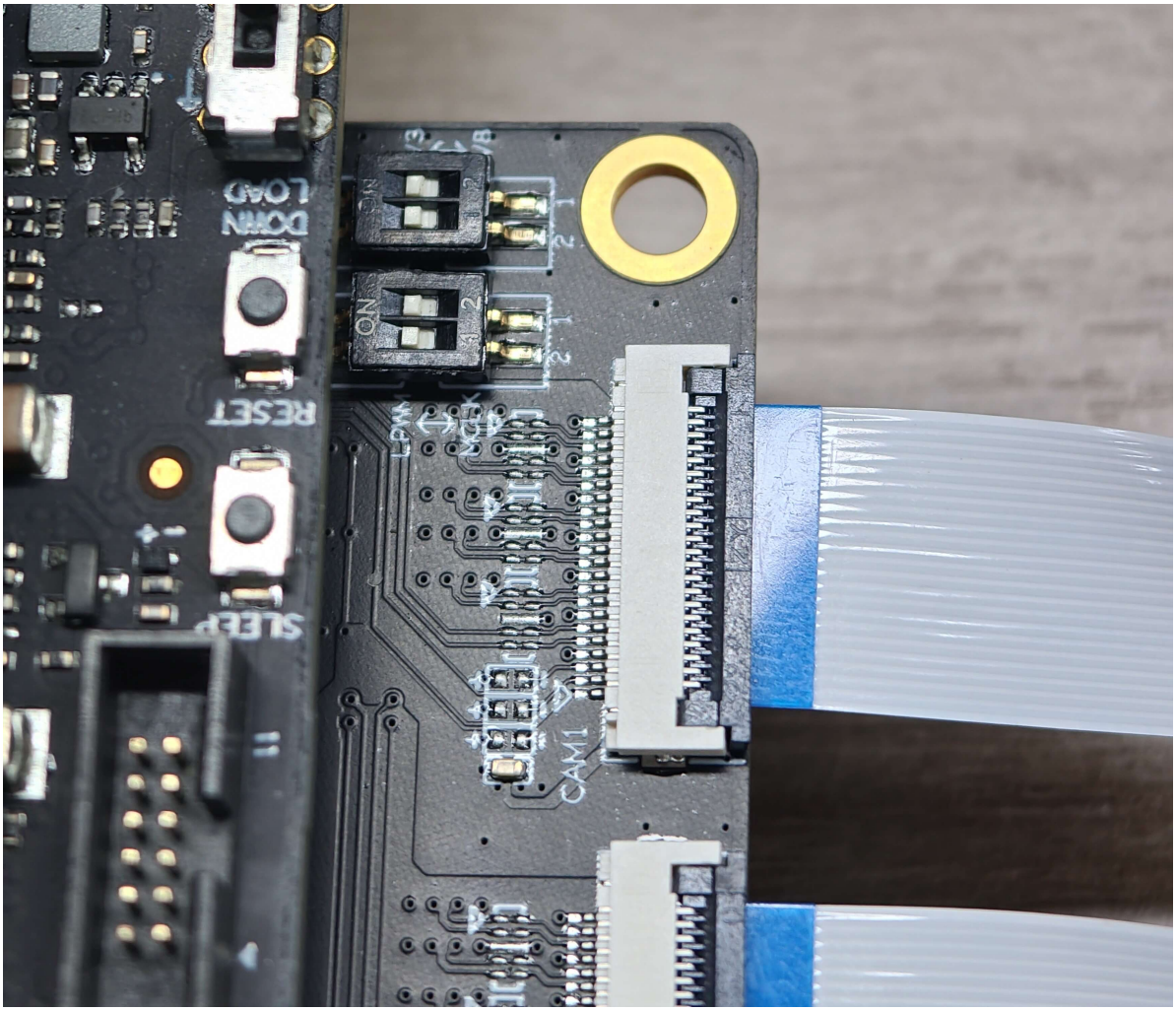
1.Confirm that the camera is correctly connected to the RDK motherboard.

RDK X5 access method

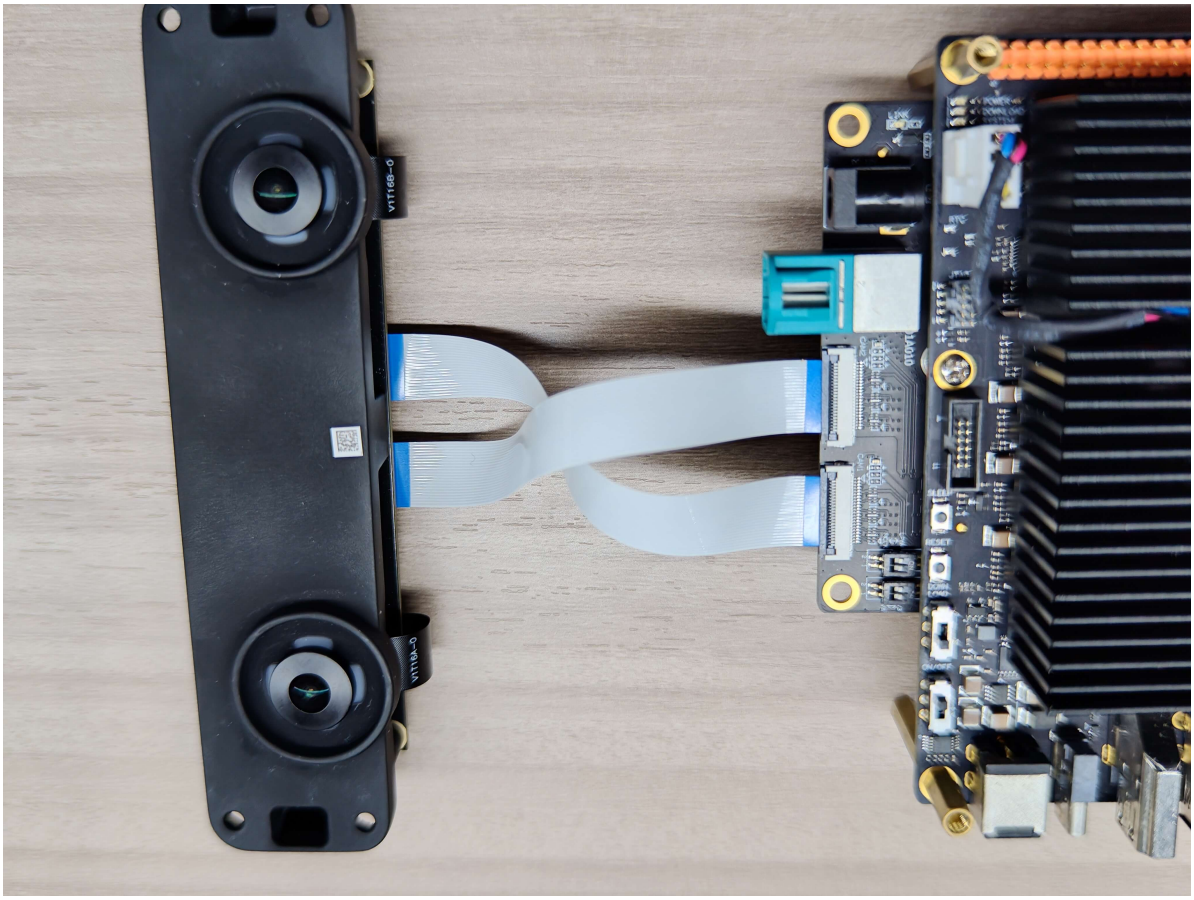


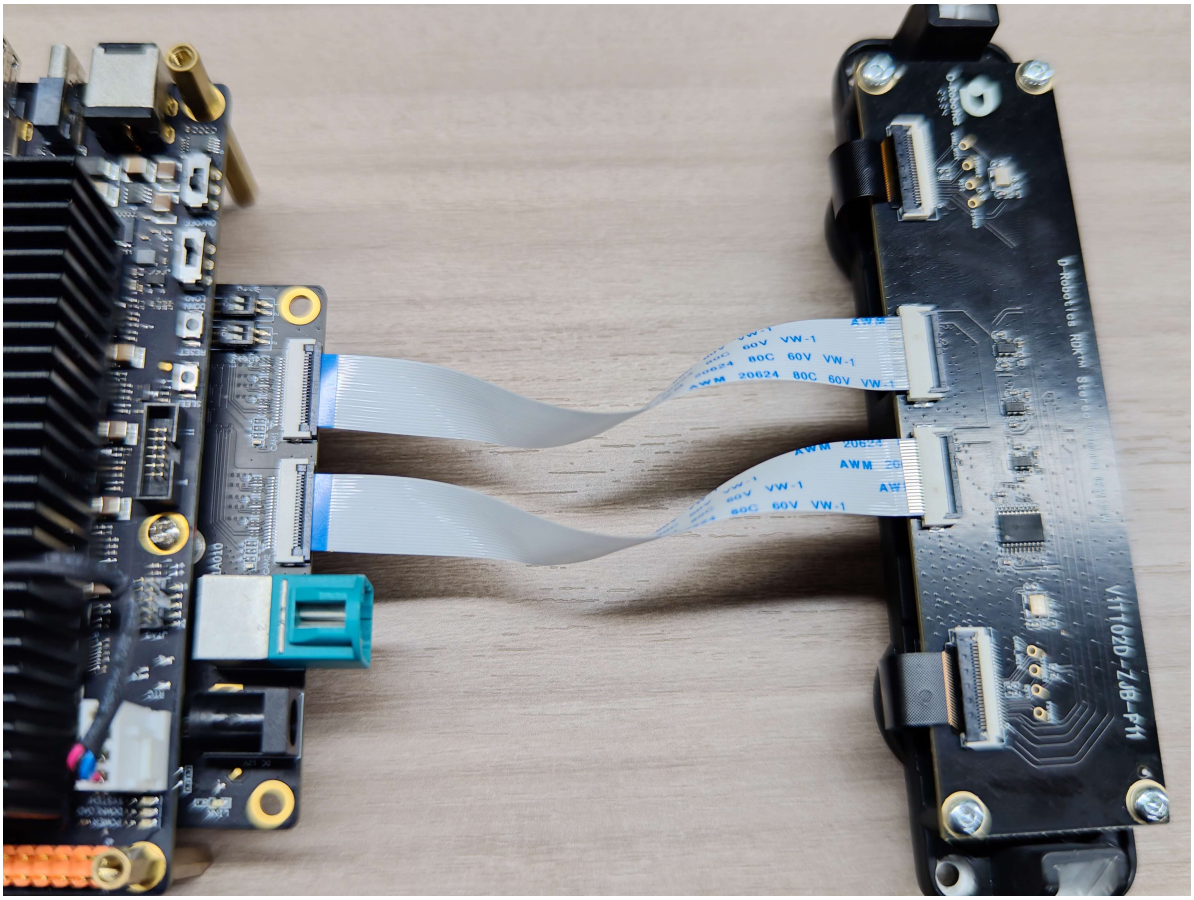
S100(P) access method

With the power off, the camera expansion board switches are set to LPWM and 3.3V respectively.



Camera wiring





2.The RDK has already burned the Ubuntu 22.04 system image.

3.RDK has successfully installed tros.b

4.Confirm that the PC can access RDK via the network.

How to use

Log in to RDK via SSH

Start the hobot_sensor node using the following command.

```
# Configure the tros.b environment
source /opt/tros/humble/setup.bash
```

```
# Launch mode
ros2 launch mipi_cam mipi_cam_dual_channel.launch.py
```

To view images from a stereo camera on the web, since the raw data needs to be published, a node encoding JPEG images and a node publishing via a web service are required. The startup command is as follows:

```
# Configure the tros.b environment
source /opt/tros/humble/setup.bash
```

```
# Launch mode
ros2 launch mipi_cam mipi_cam_dual_channel_websocket.launch.py
```

Open a browser on your PC (Chrome/Firefox/Edge) and enter `http://IP:8000` (where IP is the RDK IP address). Click on the Web display in the upper left corner to see the real-time output from the binoculars.

Precautions

1. Precautions for Camera Insertion and Removal

   It is strictly forbidden to insert or remove the camera while the development board is powered on, otherwise the camera module may be easily damaged.

2. If you encounter a startup error with the hobot_sensor node, you can troubleshoot the problem using the following steps:

   Check the hardware connection;

   Check if the tros.b environment is configured;

   Check if the parameters are correct, please refer to [README.md]() for details;